Name: _____ Period: _____

Unit 3: Artemis II Video Log

All unit long, we will watch real footage from NASA's Artemis II mission, the first time humans have traveled around the Moon in more than 50 years. Answer the questions for each video as you watch. Keep this sheet in your folder until the unit test.

Why Do Astronauts Float?

Video: Artemis II Launches Astronauts to the Moon (Official NASA Recap)
ttps://www.youtube.com/watch?v=_eeZQw9PBc0 (2:23)

1. According to the video, what makes the Artemis II crew's trip historic?

_____

2. Watch the liftoff closely. What force was the rocket working against to leave the ground? What did you see that shows how strong that force is?

_____

_____

Video: Mission Spotlight: Artemis II Zero Gravity Indicator
https://www.youtube.com/watch?v=q2YxFKBzWEQ (2:08)

3. What is the job of a zero gravity indicator like Rise?

_____

4. Why do the astronauts need a toy to tell them this, instead of noticing it themselves? (Hint: think about where the astronauts are during launch.)

_____

5. *(Answer at the end of class.)* The toy is called a "zero gravity" indicator. Based on what you learned today, is that name accurate? Explain.

_____

_____

Farther Than the Moon

Video: Artemis II Lunar Flyby News Report
https://www.youtube.com/watch?v=YxTfy9N8ock (0:04-2:44)

6. What record did the Artemis II crew set during the flyby, and whose record did they break?

_____

7. One astronaut said they would keep traveling farther "before Mother Earth succeeds in pulling us back."

a. What force is he talking about?

_____

b. What does this tell you about that force, even near the Moon?

_____

8. The crew expected to lose contact with NASA for about 40 minutes. Use where they were going to explain why. (Hint: the far side of the Moon is never visible from Earth.)

_____

9. The reporter said the crew flew “thousands of feet” above the Moon. The real distance was about 4,067 miles. Why is it important to check facts, even from the news?

_____

Video: Animation of Artemis II Flight Path
https://www.youtube.com/shorts/D1dwkxOSGYA

10. Look at the shape of the path. Where do you think Orion was moving fastest: near the Moon, at its farthest point from Earth, or right before it reached Earth? Explain.

_____

_____

Darkest Places on the Moon

Video: Artemis II and the Moon’s Shadowed Craters (watch 2:05 to 3:21)
https://www.youtube.com/watch?v=qKlu4h7a5ZE&t=3s

11. Where on the Moon are the permanently shadowed craters found?

_____

12. Why can water ice stay frozen inside these craters for a very long time?

_____

13. Even if there is ice in these craters, why might it be hard for astronauts to use it? Give one reason from the video.

_____

14. *(Answer at the end of class.)* The far side of the Moon gets sunlight, so why do some craters near the poles stay dark forever?

_____

_____

Phase or Fake? + Eclipse Over Buffalo

Video: Artemis II Flyby Photos News Report
https://www.youtube.com/watch?v=Z2JuxuJr04g 2:41

15. How long did the solar eclipse the crew saw last? What could they see glowing around the outline of the Moon?

_____

16. During this eclipse, the Sun, the Moon, and Orion were lined up. Write them in order from one end of the line to the other.

_____ → _____ → _____

17. No one on Earth saw this eclipse. Explain why only the crew could see it. (Hint: think about where the Moon’s shadow fell.)

_____

18. The crew could see planets like Mars during the eclipse. Why would planets and stars become easier to see during a solar eclipse?

_____

Make the Moon

Video: Artemis II Crew Sees Impact Flashes
https://www.youtube.com/watch?v=B9MoWCnmuqw 1:51

19. Describe what an impact flash looked like to the crew. Was it like a shooting star?

_____

20. According to the scientist, what causes an impact flash on the Moon?

_____

21. Small objects are still hitting the Moon today. What does that tell you about how the Moon got all of its craters?

_____

22. *(Answer at the end of class.)* Objects hit Earth too, but Earth has far fewer craters than the Moon. Give one reason why.

_____

Review

Video: Artemis II Re-entry and Splashdown
https://www.youtube.com/watch?v=37HxR2dfMJM 2:19

23. How fast was Integrity (the crew's name for their Orion capsule) traveling when it re-entered Earth's atmosphere?

_____

24. Why was Orion moving so fast right before it reached Earth? Use what you learned about orbits this unit.

_____

25. The video says that during re-entry, "the crew starts feeling gravity." Was gravity missing during the rest of the trip? Explain what actually changed. (Hint: think back to Rise.)

_____

_____